

01. Document Information

Field	Value
Project Name	Supply Chain Performance & Analytics Dashboard
Document Type	Business Requirements Document
Project Type	Data Analytics & Business Intelligence
Industry	Supply Chain & Retail
Role	Data Analyst
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Tools	Supabase , PostgreSQL, SQL, Power BI, DAX
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Status	Completed

02. Business Context

Supply chain performance depends on several connected areas, including procurement, inventory, production, and sales. Analyzing these areas together helps identify operational risks, performance gaps, and product-level opportunities.

This project uses operational data hosted on Supabase and SQL analysis to evaluate supply chain performance. The results are presented through an interactive Power BI dashboard for easier business analysis and decision-making.

03. Business Requirements

The project should provide analysis that helps users:

BR-01 — Procurement Performance

Evaluate supplier spending, procurement quantity, procurement cost, lead time, and quality.

BR-02 — Inventory Performance

Identify products and facilities with inventory levels at or below reorder points and highlight inventory shortfalls.

BR-03 — Production Performance

Compare production output across facilities and identify products with higher defective units.

BR-04 — Sales Performance

Analyze product revenue, sales volume, profit, and profit margin.

BR-05 — Product Performance

Identify high-performing products based on revenue, profit, sales volume, and profitability.

BR-06 — Interactive Reporting

Provide an interactive Power BI dashboard that allows users to explore supply chain performance using relevant filters.

04. Key Business Questions

The analysis should answer the following questions:

Procurement

- Which suppliers account for the highest procurement spend?
- Which products and suppliers have higher procurement costs or lead times?
- Which products have the highest procurement volumes?

Inventory

- Which product-facility combinations are currently at or below reorder points?
- Which product-facility combinations have the largest inventory shortfalls?
- Which facilities have the highest proportion of inventory records at risk?
- Which products have the highest current inventory?
- Which products have the highest estimated inventory value?

Production

- Which facilities have the highest production output?
- Which facilities have the highest production defect rates?
- Which products have the highest defective units?

Sales

- Which products generate the highest revenue?
- Which products have the highest sales volume?
- Which products generate the highest profit?
- Which products have the highest profit margins?

05. Key Performance Indicators (KPIs)

The dashboard should track the following core KPIs:

KPI	Purpose
Total Revenue	Measure overall sales revenue
Total Units Sold	Measure sales volume
Total Profit	Measure overall profitability
Profit Margin	Measure profitability relative to revenue
Total Production	Measure production output
Production Defect Rate	Monitor production quality
Current Inventory	Monitor current stock levels
Inventory Risk Rate	Identify inventory records at or below reorder point
Total Procurement Spend	Measure procurement expenditure
Total Procurement Quantity	Measure procurement volume
Average Procurement Cost	Monitor procurement cost
Average Lead Time	Monitor supplier lead time

06. Analytical Requirements

Procurement & Supplier

The analysis should evaluate:

- Procurement spend by supplier
- Procurement quantity by product
- Average procurement cost by supplier
- Supplier procurement spend and lead time

The analysis should help identify suppliers with higher spending, longer lead times, and relatively lower quality scores.

Inventory & Production

The analysis should evaluate:

- Inventory risk by facility
- Current inventory by product

- Inventory shortfalls against reorder points
- Estimated inventory value by product
- Production output by facility
- Defective units by product
- Production defect rate

Note: Estimated inventory value is an analytical estimate based on inventory quantity and average procurement cost. It is not an accounting inventory valuation.

Sales & Product Performance

The analysis should evaluate:

- Revenue by product
- Sales volume by product
- Profit by product
- Profit margin by product
- Profit by product line
- Revenue by product category

These analyses should help identify high-revenue, high-profit, high-volume, and high-margin products.

07. Dashboard Requirements

The dashboard should be organized into four pages.

Page 1 — Executive Overview

Should provide a high-level view of:

- **Total Revenue**
- **Total Units Sold**
- **Total Profit**
- **Total Production**
- **Monthly Revenue trend**
- **Top products by revenue**
- **Production output by facility**
- **Top products by profit margin**

Page 2 — Procurement & Supplier

Should provide:

- Procurement Spend
- Procurement Quantity
- Average Procurement Cost
- Average Lead Time
- Procurement spend by supplier
- Procurement quantity by product
- Average procurement cost by supplier
- Supplier procurement spend vs lead time

Page 3 — Inventory & Production Performance

The page should focus on:

- **Current Inventory**
- **Inventory Risk Rate**
- **Total Production**
- **Production Defect Rate**
- **Inventory risk by facility**
- **Current inventory by product**
- **Defective units by product**
- **Estimated inventory value by product**

Page 4 — Sales & Product Performance

The page should focus on:

- **Total Revenue**
- **Total Units Sold**
- **Total Profit**
- **Profit Margin**
- **Top 5 products by profit**
- **Profit by product line**
- **Sales volume by product**
- **Revenue by product category**

08. Filtering & Interaction Requirements

The dashboard should allow users to filter the analysis using relevant dimensions such as:

- Date
- Product
- Product Category
- Facility
- Supplier, where relevant

The filters should update the relevant dashboard visuals dynamically.

09. Expected Business Outcomes

The completed analysis should help users:

- Understand overall supply chain performance.
- Identify major procurement spending patterns.
- Detect inventory risks and potential stock shortfalls.
- Monitor production output and quality.
- Identify high-performing and profitable products.
- Compare performance across relevant suppliers, facilities, and product groups.
- Make more informed, data-driven supply chain decisions.

10. Business Acceptance Criteria

The project will be considered successful when:

- Supply chain data is analyzed using SQL.
- Data is validated before dashboard development.
- Procurement, inventory, production, and sales analysis are available.
- Core KPIs are implemented using DAX in Power BI.
- The dashboard contains four analytical pages.
- Relevant filters work across the dashboard.
- Key business questions can be answered from the analysis.
- Insights are presented clearly enough to support business decisions.

11. Data Scope

The final integrated analysis excludes the shipment dataset because its key IDs were not consistent with the available product, customer, and facility dimensions.

Capacity utilization analysis is also excluded because the available production and capacity data were not sufficiently aligned for a reliable comparison.